HYDROGEOPHYSICAL INVESTIGATION FOR BOREHOLE DEVELOPMENT, FUTA, AKURE, ONDO STATE.

CLIENT: NAME WITHELD FOR PRIVACY

CONSULTANT: MUKOLAK GEOCONSULT NIG. LTD.

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EXECUTIVE SUMMARY

A pre-drilling geophysical investigation was conducted at the Federal University of Technology, Akure (FUTA), Nigeria for groundwater development. The survey engaged the Electrical Resistivity Geophysical Method by utilizing both the Horizontal Profiling (HP) technique and Vertical Electrical Sounding (VES) technique. Two (2) traverses were established in which traverse 1 was oriented in east - west direction, while traverse 2 was in west – east direction. HP technique established, on the traverses, revealed points of anomalous resistivity values where quantitative study through the VES can be carried out. Leveraging on this information, VES points 1, 2 and 3 were located at distances 80, 105 and 160 m along traverse 1 respectively. VES 4, 5 and 6 were located at distances 45, 50 and 50 m on traverse 2. The interpretation of the VES data revealed that curve types; HA (VES 1, 2 and 3), KHA (VES 4) and H (VES 5 and 6) were delineated. Further interpretation revealed that the area is underlain by Top Soil, Weathered Layer, Fractured basement and Basement. Thus, the area is characterized by maximum of five (5) geo-electric layers with resistivity and thickness values range as follows: Top soil has a resistivity value range (106 – 367 ohm-m) with thickness (0.9 – 1.5 m), the weathered layer has a resistivity value range (43 – 175 ohm-m) with thickness (6.92 – 20 m), fracture basement of resistivity and thickness values 77 ohm-m and 16.1 m respectively and the basement having resistivity value range (323 – 3478 ohm-m) with infinite thickness**. The geoelectric parameters from the VES data suggests that, VES locations 1, 2, 5 and 6 do not show possibility of being productive for groundwater abstraction through borehole drilling. This is because the weathered layer (VES 1 and 2) is characterized with clayish formation which cannot serve as a good source of water. On the other hand, the weathered layer resistivity values (VES 5 and 6) showed evidence of saturation but characterized with shallow depth. In addition, these four VES locations are also characterized with relatively thin overburden which may not offer efficient water storage for the intended aim. However, prospective points have been delineated at VES 3 and 4. Appreciable overburden thicknesses were delineated beneath each of the two locations with evidence of saturation. However, fractured basement and thicker overburden (53 m) delineated at (VES 4) makes it a better location than VES 3 which has overburden thickness of 21 m. Consequently, it is recommended that groundwater abstraction through borehole development can be adopted at VES 3 and 4 to a depth of about 60 – 70 m (197ft – 230ft). It is also worthy of mentioning that, VES 4 productivity have been recommended to have precedence over VES 3.**

It should however be pointed out that the electrical resistivity survey simply identifies physical structures like (fractured layer) within a rock body but does not guarantee the occurrence of water in it as it could be filled with ore body for example (Like electrical conductivity material) or may be

a dry fracture. However, the electrical method has an excellent track record when it comes to finding useful quantities of water in hard rock. It should also be stated here that the quantity of water depends solely on the rock materials in which the water interacts over the years. A geophysicist is therefore not liable to either quantify and /or describe the quality of the groundwater obtained at a VES location as they are absolutely a function of nature.

1.0 INTRODUCTION

The use of surface geophysics in evaluating the groundwater condition of an area is important not only in reducing the wild cat search through drilling but in locating the best probable point in an area thereby allowing for comparison and scientific guidance to control the choice of location. Therefore, the geophysical investigation is employed to highlight the following objectives:

(i) **the geoelectric sequence/subsurface geology of the area**

(ii) **the geoelectric parameters of the area**

(iii) **the probable aquiferous units and the nature/characteristics of the aquifer**

(iii) **the depth to the aquifer**

This will assist greatly in borehole design, drilling and development of the well.

2.0 SITE DESCRIPTION

The study area is within the Federal University of Technology Akure (FUTA), Ondo State with the aim of finding reasonable drilling point. The use of groundwater is desirable for the domestic use of people due to its portability and sustainability. The data acquisition process, was done to first assess the suitability of the underlying geology. With this, traverses 1 and 2 were established in East – West and West – East direction respectively. Horizontal profiling (HP) technique was carried out to locate zones of anomalous resistivity values which were areas of relatively moderate resistivity values considering the area might be a charnokitic environment. VES 1, 2 and 3 were positioned at a distance of 80, 105 and 160 m respectively on traverse 1 while VES 4, 5 and 6 were situated at 50, 65 and 80 m respectively on traverse 2 (Figure 1).

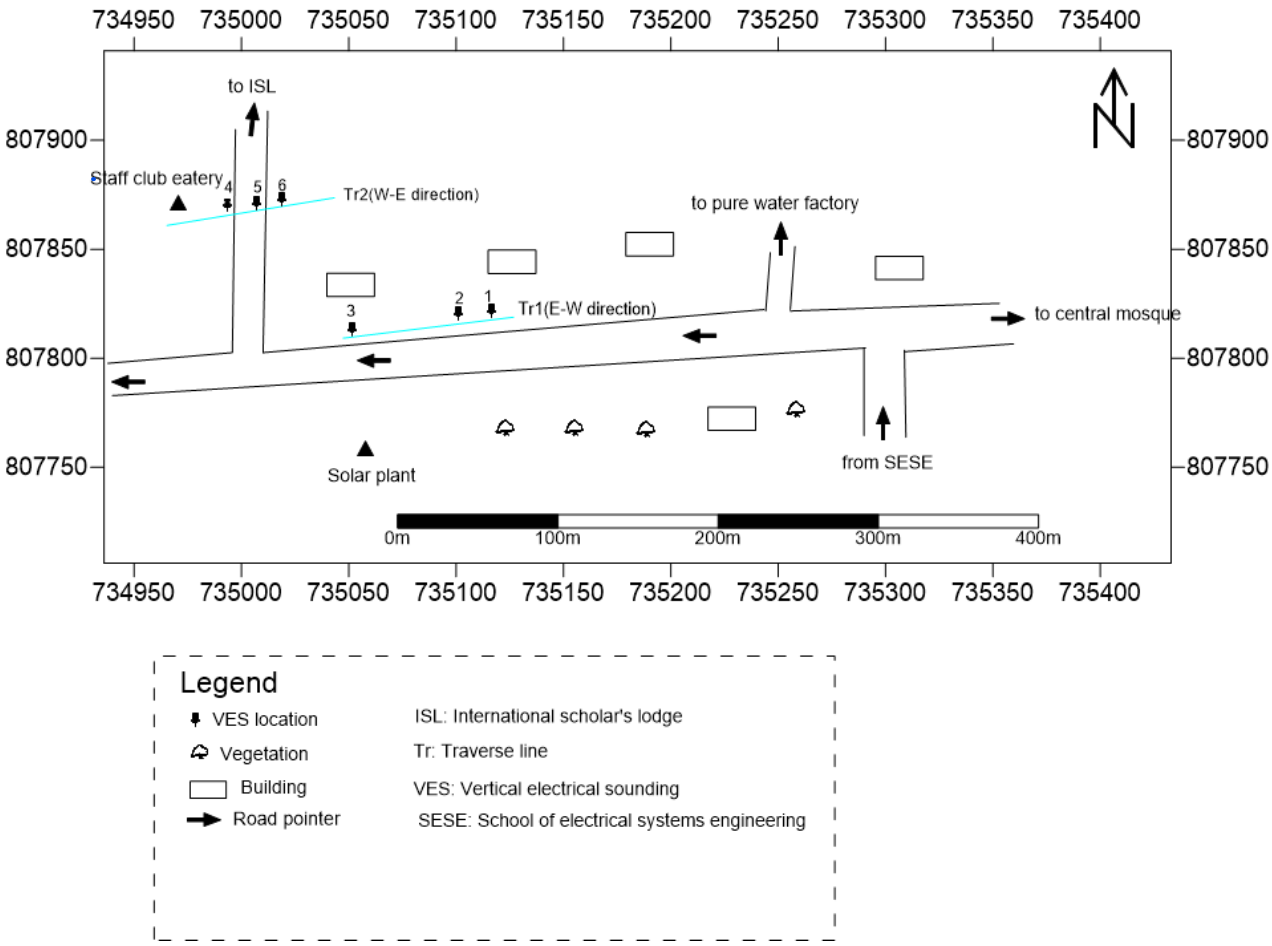

Fig. 1: Base map of the study area

3.0 GEOLOGY AND GEOMORPHOLOGY

Mainly, Precambrian basement complex rocks such as Granite have underlain the study area. The study area is situated on an undulating terrain with elevation ranges of (360 – 380 m) above the sea level. There is no visible presence of outcrops in the study area and there is presence of vegetation around the established traverse.

4.0 GEOPHYSICAL INVESTIGATION

The geophysical investigation employed both the horizontal profiling (HP) using Wenner array as well as the Vertical Electrical Sounding (VES) technique employing the normal Schlumberger configuration (Fig. 2). In the HP technique employing Wenner array, the two potential electrodes were located within the two current electrodes and an electrode spacing of 20m was adopted. Measurement were taken at each station positions which were 5m apart from each other until the whole station were exhausted within the established traverse. In the VES approach employing

Schlumberger configuration, with the potential electrodes located within the current electrodes, the half current electrode (AB/2) spreading starting from 1m and reaching a maximum spread of 150m was adopted. The R50 earth resistivity meter was used as the geophysical measuring equipment and the whole processes adopted were geared towards good and accurate data acquisition.

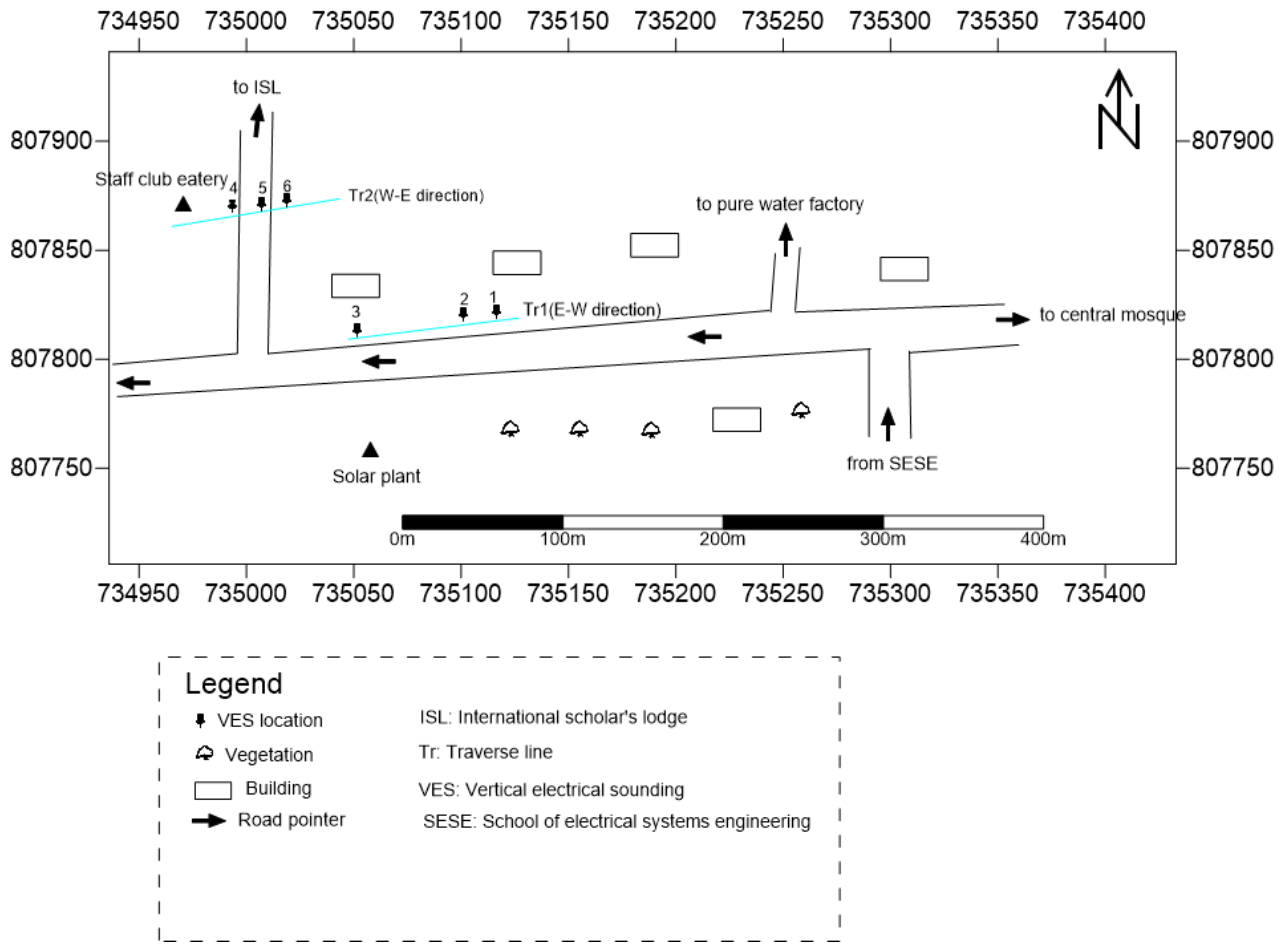

Fig. 2: Data acquisition map of the study area

5.0 DATA PRESENTATION AND INTERPRETATION

Geophysical data can be presented in the form of depth sounding curves, profiles, maps, sections and tables. For this study, the use of profile, depth sounding curves and tables in discussing and establishing the relevant subsurface layers and basement structures for the target were adopted. Figures 3a and b show the resistivity trend of the HP data along traverse 1 and 2 respectively.

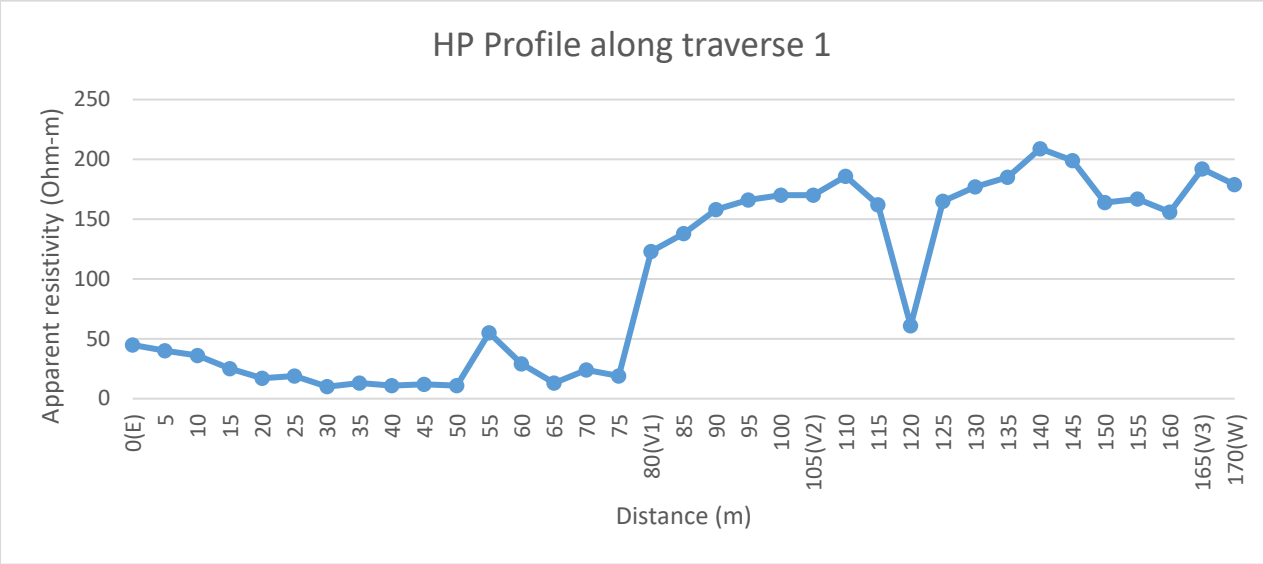

Fig. 3a: Horizontal profiling along traverse 1

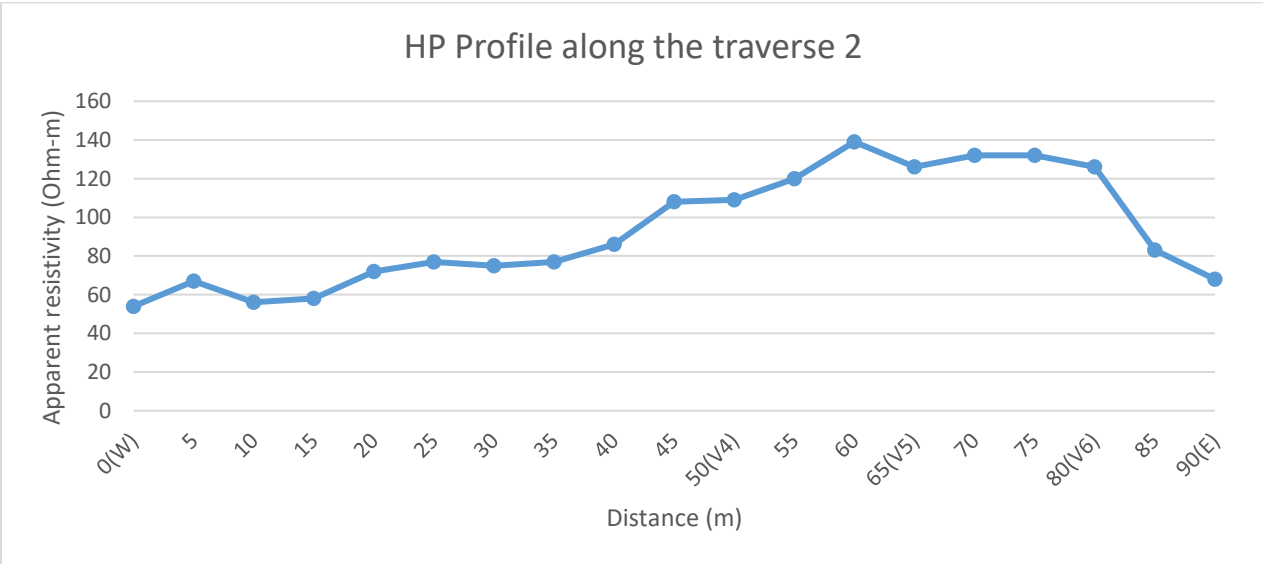

Fig. 3b: Horizontal profiling along traverse 2

Figures 3a and b elaborates the resistivity trend of the horizontal profile (HP) along traverse 1 (tr1) and traverse 2 (tr2). While tr1 was oriented in E-W direction, tr2 was adopted in W-E direction to capture the underlying geology. The result of the profile showed anomalous resistivity values (moderate values; greater than 100 and less than 200 ohm-m) along the established traverses and

could be diagnostic of saturated zones. The points were subsequently picked for further depth investigation employing the Schlumberger configured Vertical Electrical Sounding (VES) technique. VES 1, 2 and 3 were located on 80, 105 and 160 m on tr1 while VES 4, 5 and 6 were located on 50, 65 and 80 m along tr2.

The sounding data are plotted and presented as sounding curves (Figs.4a, b, c, d, e and f) and are interpreted quantitatively. The results of the VES quantitative interpretation are shown in Tables 3a through f as geoelectric parameters.

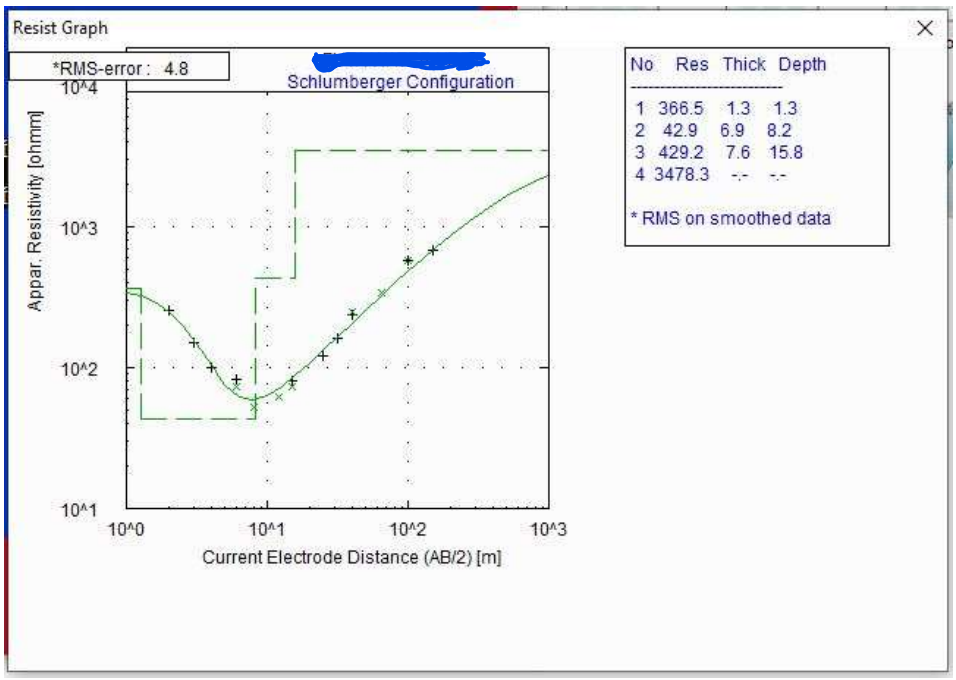

Fig. 4a: Depth sounding curve for VES 1

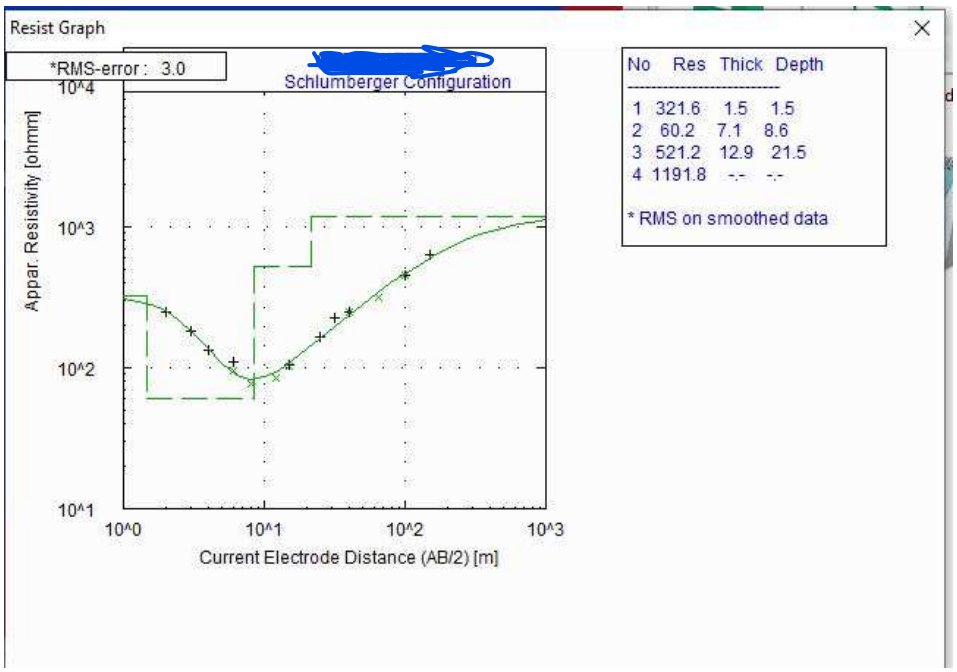

Fig. 4b: Depth sounding curve for VES 2

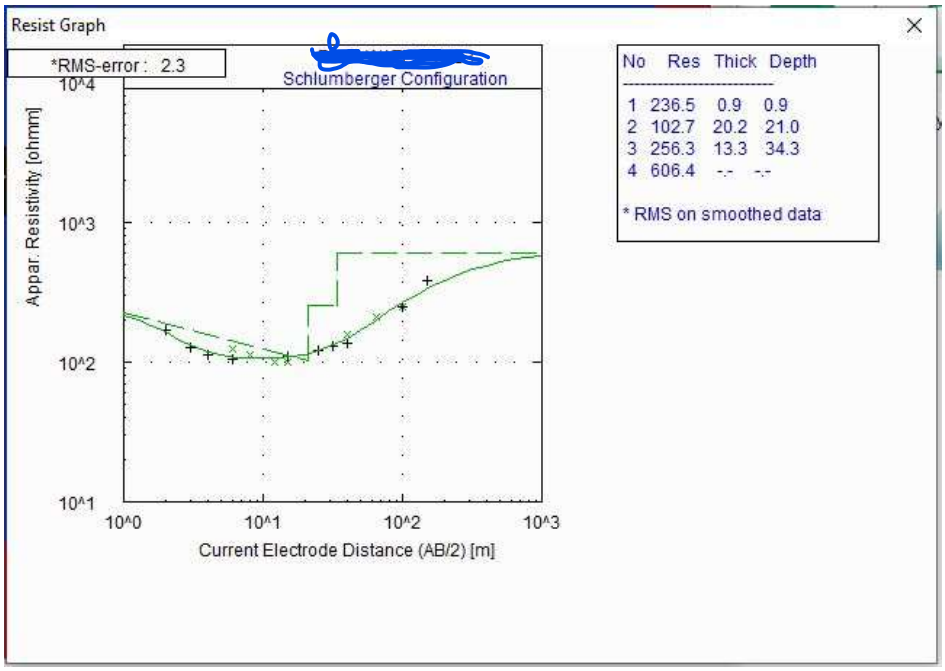

Fig. 4c: Depth sounding curve for VES 3

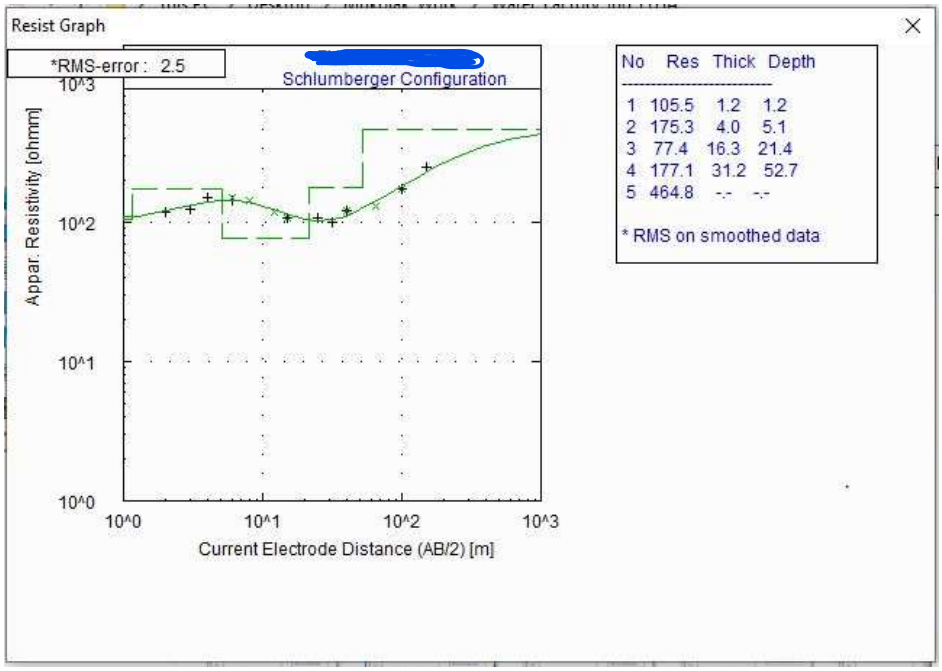

Fig. 4d: Depth sounding curve for VES 4

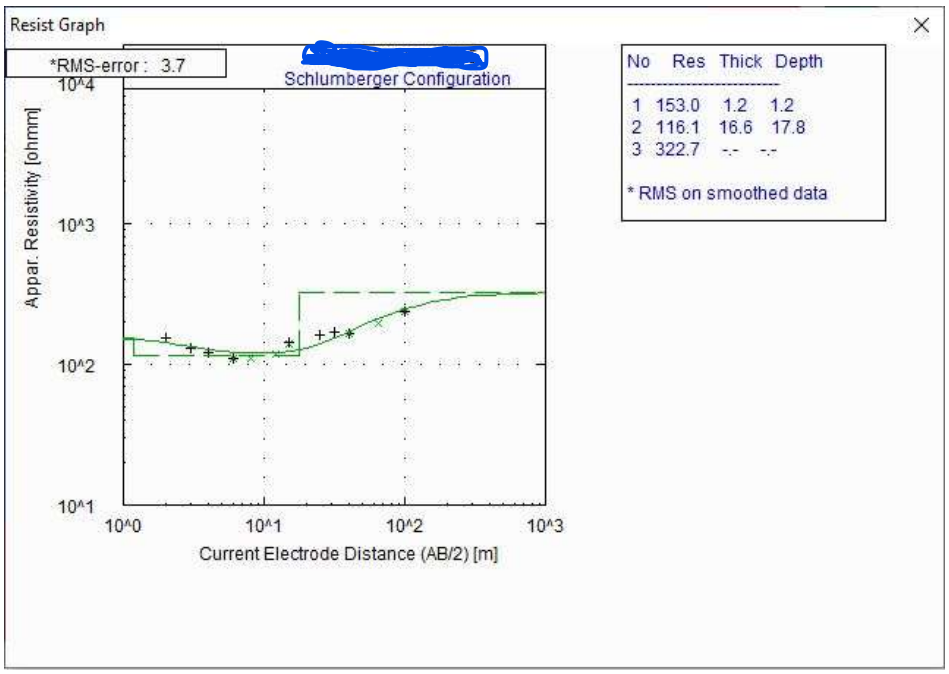

Fig. 4e: Depth sounding curve for VES 5

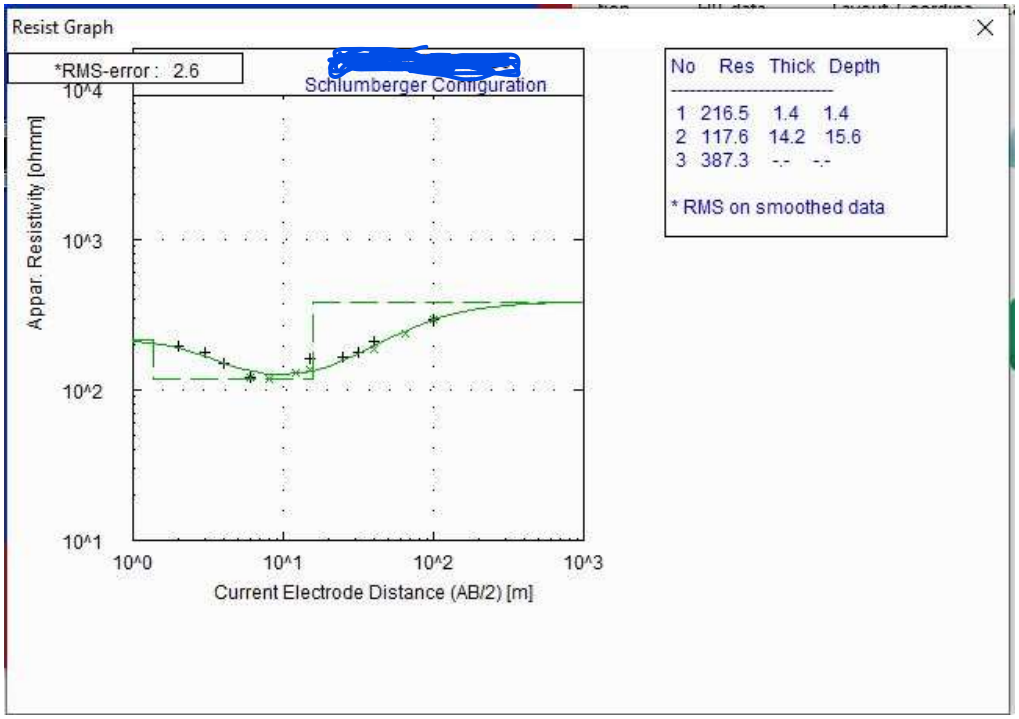

Fig. 4f: Depth sounding curve for VES 6

Table 3a: VES Interpreted Results (V1)

S/N	Resistivity (Ωm)	Thickness (m)	Depth (m)	Lithologic Equivalence	Curve type
1	367	1.3	1.3	Topsoil	HA
2	43	6.9	8.2	Weathered Layer	
3	429	7.6	15.8	Partially Weathered Layer	
4	3478	-	-	Fresh Basement	

Table 3b: VES Interpreted Results (V2)

S/N	Resistivity (Ωm)	Thickness (m)	Depth (m)	Lithologic Equivalence	Curve type
1	362	1.5	1.5	Topsoil	HA
2	60	7.1	8.6	Weathered Layer	
3	521	12.9	21.5	Partially Weathered Layer	
4	1192	-	-	Basement	

Table 3c: VES Interpreted Results (V3)

S/N	Resistivity (Ωm)	Thickness (m)	Depth (m)	Lithologic Equivalence	Curve type
1	237	0.9	0.9	Topsoil	HA
2	103	20	21	Weathered Layer	
3	256	13.3	34.3	Partially Weathered Layer	
4	606	-	-	Basement	

Table 3d: VES Interpreted Results (V4)

S/N	Resistivity (Ωm)	Thickness (m)	Depth (m)	Lithologic Equivalence	Curve type
1	106	1.2	1.2	Topsoil	KHA
2	175	4	5.1	Weathered layer	
3	77	16.3	21.4	Fractured basement	
4	177	31.2	52.7	Partially weathered layer	
5	465	--	--	Basement	

Table 3e: VES Interpreted Results (V5)

S/N	Resistivity (Ωm)	Thickness (m)	Depth (m)	Lithologic Equivalence	Curve type
1	153	1.2	1.2	Topsoil	H
2	116	16.6	17.8	Weathered Layer	
3	323	--	--	Basement	

Table 3f: VES Interpreted Results (V6)

S/N	Resistivity (Ωm)	Thickness (m)	Depth (m)	Lithologic Equivalence	Curve type
1	217	1.4	1.4	Topsoil	H
2	118	14.2	15.6	Weathered Layer	
3	387	--	--	Basement	

From Tables 3a through f, the geo-electric parameters obtained from depth sounding classified the area into three (VES 5 and 6), four (VES 1, 2 and 3) and five (VES 4) geo-electric layers. Further interpretation revealed that the area is underlain by Top Soil, Weathered Layer, Fractured basement and Basement. Thus, the area is characterized by maximum of five (5) geo-electric layers with resistivity and thickness values range as follows: Top soil has a resistivity value range (106 – 367 ohm-m) with thickness (0.9 – 1.5 m), the weathered layer has a resistivity value range (43 – 175 ohm-m) with thickness (6.92 – 20 m), fracture basement of resistivity and thickness values 77 ohm-m and 16.1 m respectively and the basement having resistivity value range (323 – 3478 ohm-m) with infinite thickness**. The geoelectric parameters from the VES data suggests that, VES locations 1, 2, 5 and 6 do not show possibility of being productive for groundwater abstraction through borehole drilling. This is because the weathered layer (VES 1 and 2) is characterized with clayish formation which cannot serve as a good source of water. On the other hand, the weathered layer resistivity values (VES 5 and 6) showed evidence of saturation but characterized with shallow depth. In addition, these four VES locations are also characterized with relatively thin overburden which may not offer efficient water storage for the intended aim. However, prospective points have been delineated at VES 3**

and 4. Appreciable overburden thicknesses were delineated beneath each of the two locations with evidence of saturation. However, fractured basements and thicker overburden (53 m) delineated at (VES 4) makes it a better location than VES 3 which has overburden thickness of 21 m. Consequently, it is recommended that groundwater abstraction through borehole development can be adopted at VES 3 and 4 to a depth of about 60 – 70 m (197ft – 230ft). It is also worthy of mentioning that, VES 4 productivity have been recommended to have precedence over VES 3.

GROUNDWATER POTENTIAL EVALUATION

In a typical basement terrain, the groundwater prospect is hinged on a fairly thick overburden and weathered/fractured layer with resistivity parameters compatible with groundwater saturation. The weathered layer (VES 3) showed support for groundwater abstraction but has relatively thinner overburden. At VES 4, the presence of the fractured layer coupled with relatively thicker overburden makes it to have precedence for groundwater abstraction over VES 3 though the two locations are recommended. **In view of this, VES 3 and 4 are suitable for groundwater development through borehole development with VES 4 being the better option.**

CONCLUSION

The most promising lithologic unit for groundwater occurrence within the study area is the weathered/fractured layer which has relatively appreciable thickness as well as efficient overburden thickness. While at VES locations 3, the weathered layer is suspected to have resistivity value compatible with groundwater saturation and therefore can be harnessed for groundwater abstraction, VES 4 also has fractured layer offering such prospect Furthermore, VES 4 stands to be highly preferred than VES 3 due to its relatively thicker overburden. Also, VES location 4 is being underlain by a basement which is suspected to be saturated and could be drilled to a reasonable amount.

RECOMMENDATION

It is recommended that groundwater abstraction through borehole development can be adopted at VES 3 and 4 to a depth of about 60 – 70 m (197ft – 230ft). It is also worthy of mentioning that, VES 4 productivity have been recommended to have precedence over VES 3.